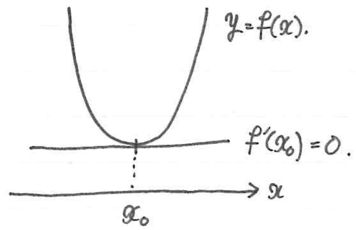
Chapter 2, Minimizing Variational problems for functionals and Laplace eq.

Let $f:\mathbb{R}\to\mathbb{R}$ be smooth func. and let $x_0\in\mathbb{R}$
be a minimzier of f $\min_{x\in\mathbb{R}} f = \inf_{x\in\mathbb{R}} f = f(x_0)$.

Then $f'(x_0)=0$.



$$f'(x_0) = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} f(x_0+\varepsilon h).\quad \forall h\in\mathbb{R}.$$

one variable. multi-variable

Let $f:\mathbb{R}^n\to\mathbb{R}$ be a smooth func. and let $x_0\in\mathbb{R}^n$
be a minimizer of f.
Then

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} f(x_0+\varepsilon h)=0,\quad \forall h\in\mathbb{R}^n.$$

$\mathbb{R}^n \to X$

infinity dim. vec. sp) analysis

or

manifold

or

surface etc...) geometry

⟨derivation of 1-dim diff eq⟩

for smooth func $u:[0,1]\to\mathbb{R}$,

let

(2.1). $E[u] := \frac{1}{2}\int_0^1 u_x^2\,dx$, $u(0)=a\in\mathbb{R}$, $u(1)=b\in\mathbb{R}$. fixed.

where $u_x(x) = \frac{d}{dx}u(x)$.

let $h:[0,1]\to\mathbb{R}$ be a smooth func. with $h(0)=h(1)=0$.

let u be minimizer of E.

◎ We want to know the properties of u.

$\to \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} E[u+\varepsilon h] = 0.$ (2.2).

$u+\varepsilon h \to u(x)+\varepsilon h(x).$

$\hookrightarrow E[u+\varepsilon h] = \frac{1}{2}\int_0^1 \left((u+\varepsilon h)_x(x)\right)^2 dx.$

$$= \frac{1}{2}\int_0^1 \left(u_x(x)+\varepsilon h_x(x)\right)^2 dx$$

$$= \frac{1}{2}\int_0^1 \left(u_x^2(x)+2\varepsilon u_x h_x(x)+\varepsilon^2 h_x^2(x)\right) dx$$

$$\Rightarrow \frac{d}{d\varepsilon} E[u+\varepsilon h] = \int_0^1 u_x(x)h_x(x)\,dx + \varepsilon\int_0^1 h_x^2(x)\,dx$$

$$\Rightarrow \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} E[u+\varepsilon h] = \int_0^1 u_x(x)h_x(x)\,dx.$$

From (2.2) $\int_0^1 u_x(x)\,h_x(x)\,dx = 0$

By integrate by parts.

$$\int_0^1 u_x(x)\,h_x(x)\,dx = \Big[\,u_x(x)\,\cancel{h(x)}^{\,0}\Big]_0^1 - \int_0^1 u_{xx}(x)\,h(x)\,dx$$
$$= -\int_0^1 u_{xx}(x)\,h(x)\,dx .$$

$\therefore\ -\int_0^1 u_{xx}(x)\cdot h(x)\,dx = 0.$

for $\forall$ func. h with $h(0) = h(1) = 0.$

Because h is arbitrary difficult to prove.
$-u_{xx}(x) = 0$, $x \in (0,1)$.

<derivation of PDEs>

let $\Omega \subset \mathbb{R}^n$ be an open set. with smooth bdd .
For any func. $u : \Omega \to \mathbb{R}$ define.

$$E[u] = \frac{1}{2}\int_{\Omega} |\nabla u(x)|^2\,dx .$$

Notation
$|\nabla u(x)|^2 = u_{x_1}^2 + u_{x_2}^2 + \cdots + u_{x_n}^2$, $x = (x_1, \dots, x_n) \in \Omega$

For given func. $\phi : \partial\Omega \to \mathbb{R}$, we seek the cond that

- u is the minimizer of E.
- $u(x) = \phi(x)$, $x \in \partial\Omega$

Let $h : \Omega \to \mathbb{R}$ be a smooth func. with $h(x) = 0$.
for $x \in \partial\Omega$.
Then
$(u + \varepsilon h)(x) = u(x) + \varepsilon h(x) = u(x) = \phi(x)$, $x \in \partial\Omega$.

$$\Rightarrow \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} E[u+\varepsilon h] = 0 \qquad (2.5)$$

same as (2.2)

Compute $E[u+\varepsilon h]$

$$E[u+\varepsilon h] = \frac{1}{2}\int_{\Omega} |\nabla(u+\varepsilon h)(x)|^2\,dx$$
$$= \frac{1}{2}\int_{\Omega} |\nabla u(x) + \varepsilon \nabla h(x)|^2\,dx$$
$$= \frac{1}{2}\int_{\Omega} \big(\nabla u(x) + \varepsilon\nabla h(x)\big)\big(\nabla u(x) + \varepsilon\nabla h(x)\big)\,dx$$
$$= \frac{1}{2}\int_{\Omega} \big(|\nabla u(x)|^2 + 2\varepsilon\nabla u(x)\cdot\nabla h(x) + \varepsilon^2|\nabla h(x)|^2\big)\,dx$$

$$\Rightarrow \frac{d}{d\varepsilon} E[u+\varepsilon h] = \int_{\Omega} \nabla u(x)\cdot\nabla h(x)\,dx + \varepsilon\int_{\Omega} |\nabla h(x)|^2\,dx$$

$$\therefore \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} E[u+\varepsilon h] = \int_{\Omega} \nabla u(x)\cdot\nabla h(x)\,dx$$ for $\forall$ func. h with $h|_{\partial\Omega} = 0$.

$$(2.5) \Rightarrow \int_{\Omega} \nabla u(x)\cdot\nabla h(x)\,dx = 0.$$

Recall div thm.

$$\int_\Omega \operatorname{div} \vec{F}(x)\,dx = \int_{\partial\Omega} \vec{F}\cdot\vec{\nu}\,d\sigma$$

Excercize: For any scalar field f, g

$$\int_\Omega \nabla f(x)\cdot\nabla g(x)\,dx = \int_{\partial\Omega} g\,\nabla f(x)\cdot\vec{\nu}(x)\,d\sigma - \int_\Omega g\cdot\operatorname{div}\nabla f(x)\,dx$$

Using dim thm (integration of parts).

$$\int_\Omega \nabla u(x)\cdot\nabla h(x)\,dx$$

$$= \int_{\partial\Omega} \underbrace{h(x)}_{\overset{\shortparallel}{0}}\cdot\nabla u(x)\cdot\vec{\nu}(x)\,d\sigma - \int_\Omega \operatorname{div}(\nabla u(x))\,h(x)\,dx$$

$$= \int_\Omega (-\operatorname{div}(\nabla u(x)))\cdot h(x)\,dx$$

Excercize

$$= \int_\Omega (-\Delta u(x))\cdot h(x)\,dx$$

Because h is arbitrary
$-\Delta u(x) = 0,\quad x\in\Omega$
(Laplace eq).

How to guarantee / derive?

For conti func. $h:\Omega\to\mathbb{R}$ define.
~~cpt~~
$\operatorname{spt} h := \overline{\{x\in\Omega\,;\,h(x)\neq 0\}}$

spt h is called support of h

Define

theory of metric sp. general topology

$C_0^\infty(\Omega) := \{h:\Omega\to\mathbb{R},\ C^\infty$
spt h is cpt relatively in $\Omega\}$

spt $h\subset\Omega$ and spt h is b'dd.

Thm 2.1 (Fundamental lem. for Calculus of variation).
Let $f:\Omega\to\mathbb{R}$ be a locally integrable func. on an open set Ω
Assume for any $h\in C_0^\infty(\Omega)$.

$$\int_\Omega f(x)\cdot h(x)\,dx = 0.$$

Then $f(x)=0$. for almost all $x\in\Omega$ ▨

Example:

$$f(x) = \begin{cases} 1 & (x=0) \\ 0 & (x\in\mathbb{R}\setminus\{0\}) \end{cases}$$

$\Rightarrow$ In Thm 2.1 it is impossible to change "almost all" to "all". But if f is conti. then "almost" $\to$ "all".
$\Downarrow$
We can obtain Laplace eq.

Remark: We used

$$\underbrace{\int_\Omega \nabla u(x)\cdot\nabla h(x)\,dx}_{\text{1st-derivative}} = \underbrace{\int_\Omega (-\Delta u(x))\cdot h(x)\,dx}_{\text{2nd-derivative}}$$

1st-derivative (weak). FEM, Galerkin

2nd-derivative (storonge, classical).